

## SKILLS

**Languages:** Python, C++, JavaScript/TypeScript, SQL, Swift, Bash

**Software Engineering:** REST API design, microservices, distributed systems, system design, fullstack development (React, React Native, Next.js, Node.js, FastAPI)

**Machine Learning & AI:** LLMs, RAG, Vision-Language Models, Whisper ASR, YOLO, MediaPipe, PyTorch, TensorFlow, XGBoost/LightGBM, model evaluation

**Cloud & Data:** AWS (EC2/S3/ElastiCache), Docker, MySQL, Redis, Git/GitHub, CI/CD-ready workflows

## EXPERIENCE

### Transnode AI, LLC | Software Engineer Intern

Jun 2026 – Aug 2026

- Owned **customer-facing features** for Earnie end-to-end across **React Native, iOS, and web**, moving fast from design through implementation to launch while resolving integration and release-blocking issues.
- Leveraged **AI-assisted development** tools to accelerate implementation while critically validating every output, contributing to Earnie's registered user base growing **~111%** (679 to 1,436 users) since June 2026.

### Bharati Vidyapeeth COE | ML Research Assistant

Jan 2024 – Jun 2024

- Designed and built a **RAG pipeline** (Python, PyTorch, LLMs) grounding generated responses in structured knowledge and semantic retrieval, applying rigorous performance profiling to identify and resolve bottlenecks.
- Applied controlled experimentation to evaluate system quality and real-time responsiveness, cutting end-to-end response latency **65%** with measurable, reproducible results.

### Redis Ltd | Software Engineer Intern

Jun 2023 – Sep 2023

- Designed, built, and tested **Redis 7.2 caching** workflows for distributed systems handling large-scale, high-throughput, low-latency data access on **AWS ElastiCache**, cutting memory usage **18%**.
- Automated validation and regression testing, improving testing efficiency **30%** while strengthening system reliability at scale.

## PROJECTS

### Sociopals

github.com/RuchirBhatia4/sociopals

- Independently designed and shipped a **fullstack real-time platform** solo end-to-end (Next.js, TypeScript, Node.js, Prisma/MySQL, Socket.io), owning every decision from data model to UI with no existing template to build from.
- Built production **authentication, access control, real-time messaging, and audio/video channels** from scratch, architecting reusable APIs and relational data models.

### Multimodal In-Vehicle AI Assistant

github.com/RuchirBhatia4/Multimodal-In-Vehicle-AI-Assistant

- Built an **on-device co-pilot**: YOLO11 + ByteTrack perception, driver monitoring, and a **4-bit VLM** tool-calling agent.
- Trained an **eye-state model** (blink F1 **0.95**, false alerts **33/h→0**); drowsiness alert caught **10/12 unseen** drowsy drivers.

### PitWall AI

pitwall-ai-f1.vercel.app | github.com/RuchirBhatia4/Pitwall-AI

- Built and deployed a **live F1 race-strategy engine** (FastAPI, Next.js): a **Bayesian tyre-wear model** updated lap-by-lap from live timing, an exact **dynamic-programming** optimiser over 0–3 stop plans, and **Monte Carlo** safety-car risk.
- Validated **walk-forward** on 173 driver-races, beating a naive baseline on first-stop timing (**9.2 vs 10.3** laps error) and stop count (**33.5% vs 27.7%**); live pit calls in under 3 s.

## PUBLICATIONS

### IEEE Published Paper – OS Scheduling & Page Replacement

Research Paper

- Peer-reviewed analysis of page-replacement and CPU scheduling algorithms for OS memory management.

### Patent Accepted – LLM Healthcare Chatbot

Patent

- RAG-based healthcare chatbot combining retrieval-augmented generation with structured medical knowledge for clinical QA.

### Research Paper – VLM4Rec: Multimodal Recommendation

Research Paper

- Grounded item images into semantic descriptions via a vision-language model, outperforming fusion-based baselines.

## HONORS AND AWARDS

- Registered Software Copyright** for Page-Replacement Algorithms using Shell Script. (Copyright)
- Top 5% of B.Tech Computer Engineering Class** at Bharati Vidyapeeth College of Engineering.
- 4.0 GPA & Joint Top Scorer**, DSCI 552 and Information Retrieval coursework, USC M.S. in Computer Science.

## EDUCATION

### M.S., Computer Science, University of Southern California

May 2027

- Coursework: CSCI 566: Deep Learning and its Applications, CSCI 570: Analysis of Algorithms, CSCI 642: Robot Learning

### B.Tech, Computer Engineering, Bharati Vidyapeeth College of Engineering

Jun 2024

- GPA: 3.87/4.00 (Top 5% of class) • Coursework: Algorithm Design and Analysis, Machine Learning